

Mechanism Design Audit of Comparable-Based Dynamic Fees (Shielded Labs)

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Process note. This audit is intended to surface incentive trade-offs and failure modes in an evolving fee-market proposal. It should be read as a working document to support iteration, not as a final verdict.

Abstract

Shielded Labs proposes a comparable-based dynamic fee mechanism for Zcash that (i) derives a recommended marginal fee from recent blocks, (ii) intentionally coarsens the fee space (powers of ten) for privacy and user simplicity, and (iii) keeps consensus changes minimal by pushing most logic into wallet/node policy. This report models the mechanism as a repeated game between users, relay policy, and miners; evaluates equilibrium behavior, manipulation incentives, and spam-resilience; and provides spec-level recommendations.

1 Executive Summary

Shielded Labs’ proposal can be understood as an **algorithmic posted-price estimator with a discrete fee alphabet** (powers of ten), plus an optional **priority tier** (“fast lane”) at $10\times$ the recommended marginal fee. It aims to preserve Zcash’s privacy values by reducing fee entropy and to keep consensus changes minimal by starting with wallet/node policy.

Key takeaways.

- **Policy convergence is part of the mechanism.** In a policy-first rollout, the estimator does not bind unless relay policy and miner templates converge; otherwise the market reverts to heterogeneous first-price heuristics.
- **Anti-spam is dominated by the fee floor and any utilization backstop.** In low demand, the estimator intentionally pins to the floor, so the floor is a security parameter, not a tuning knob.
- **Congestion does not automatically raise price.** Fees rise only if *included* transactions pay more, so a sustained full-block “floor-fee” regime is possible without an explicit utilization-driven rule.
- **Coarse bucketing improves privacy but introduces cliff effects.** Powers-of-ten rounding creates $10\times$ discontinuities and sharp phase transitions around bucket boundaries.

- **Fast-lane semantics are underspecified.** Under ZIP-317’s capped weighting, paying $10\times$ does not guarantee $10\times$ priority; without a standardized selection rule, the mechanism can revert to first-price behavior during congestion.
- **Median-of-medians helps manipulation resistance, and fee burning helps more.** Medians reduce outlier impact; if a meaningful fraction of fees is burned (ZIP-235-style), miner “wash fee” manipulation becomes genuinely costly.

Overall assessment. The design direction is promising for a privacy coin: it deliberately trades allocative efficiency for low fee entropy and deployability (policy-first). The central open question is whether the mechanism can guarantee that sustained congestion makes griefing expensive *without* importing an EIP-1559-like utilization rule into consensus or policy.

Qualitative scorecard.

Criterion	Assessment
Privacy (fee entropy)	Strong
Deployability (policy-first)	Strong
Spam resistance	Depends on f and backstop
Congestion response	Discontinuous / lagged
Miner manipulation resistance	Moderate; improved by burning
Equilibrium concept	Learning/CCE more apt than ESS

2 Scope, Assumptions, and Sources

In scope. Incentives of users, miners, and relay policy under the proposed estimator and constraints; stability (game-theoretic) of fee bidding; spam-resilience; miner manipulation; and privacy-related mechanism constraints (fee entropy).

Out of scope. Cryptographic privacy of Zcash, implementation bugs in any specific node or wallet, and detailed quantitative calibration (which requires simulation and empirical measurement).

Assumptions (economic). Unless stated otherwise, we assume miners prefer higher expected fee revenue subject to protocol rules; users/wallets follow simple adaptive policies rather than solving full-information equilibria; and relay policy is sufficiently convergent that “recommended” fees are not systematically undercut by alternative mempool policies.

Primary sources.

- Shielded Labs Dynamic Fees page: <https://shieldedlabs.net/fees/>.
- Zcash Dynamic Fees Lab (including “Our Proposal”): <https://fees.shieldedinfra.net/>.
- Reference implementation used by the lab: no public repository found as of February 2026 (earlier references to `ShieldedLabs/fees` resolve to 404).
- ZIP-317 (action-based fees and conventional fee definition): <https://zips.z.cash/zip-0317>.
- ZIP-313 (background on fixed fees and DoS): <https://zips.z.cash/zip-0313>.

- ZIP-235 (fee burning via the NSM): <https://zips.z.cash/zip-0235>.
- EIP-1559 (comparison point): <https://eips.ethereum.org/EIPS/eip-1559>.

3 Literature Review and Key Insights

Scope note. This section is intentionally selective (not exhaustive) and focuses on results that materially change how we should evaluate this proposal.

Curation note (why some papers moved across drafts). Earlier draft iterations temporarily removed some papers when links were unstable, duplicate, or only tangentially connected to this mechanism’s design levers. In this version, Ferreira-style dynamic posted-price work is retained because the proposal is a posted-price estimator; Gafni-style discrete/Bayesian fee work is retained because the proposal enforces a discrete fee alphabet; and Chitra/Angeris resource-pricing work is added because it sharpens how we specify “what resource is being priced.” Papers focused primarily on redistribution or MEV-internal auction plumbing are cited only when they change conclusions about this mechanism’s security, welfare, or policy assumptions.

Representative papers used as anchors.

- Roughgarden, *Transaction Fee Mechanism Design*. Free preprint: <https://arxiv.org/pdf/2106.01340.pdf>.
- Roughgarden, *Transaction Fee Mechanism Design for the Ethereum Blockchain: An Economic Analysis of EIP-1559*. Free preprint: <https://arxiv.org/pdf/2012.00854.pdf>.
- Roughgarden, *Transaction Fee Mechanism Design in a Post-MEV World*. Free preprint: <https://eprint.iacr.org/2024/331.pdf>.
- Ferreira, Moroz, Parkes, Stern, *Dynamic Posted-Price Mechanisms for the Blockchain Transaction-Fee Market*. Free preprint: <https://arxiv.org/pdf/2103.14144.pdf>.
- Leonardos, Monnot, Reijsbergen, Skoulakis, Piliouras, *Dynamical Analysis of the EIP-1559 Ethereum Fee Market*. Free preprint: <https://arxiv.org/pdf/2102.10567.pdf>.
- Diamandis, Evans, Chitra, Angeris, *Designing Multidimensional Blockchain Fee Markets*. Open-access proceedings page: <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.AFT.2023.4>.
- Evans, Angeris, Chitra, Diamandis, *Dynamic Pricing for Non-fungible Resources: Optimal Fees and Worst-case Guarantees for the EIP-1559 Burn Mechanism*. Free preprint: <https://arxiv.org/pdf/2208.07919.pdf>.
- Angeris, Diamandis, Moallemi, *Multidimensional Blockchain Fees Are (Essentially) Optimal*. Free preprint: <https://arxiv.org/pdf/2402.08661.pdf>.
- Carlsten, Kalodner, Weinberg, Narayanan, *On the Instability of Bitcoin Without the Block Reward*. Free version: <https://arxiv.org/pdf/1606.01996.pdf>.
- Gafni and Yaish, *Discrete & Bayesian Transaction Fee Mechanisms*. Free preprint: <https://arxiv.org/pdf/2210.07793.pdf>.

- Gafni, Yaish, Chung, Ferreira, *Barrier to Collusion-Resistant Transaction Fee Mechanisms*. Free preprint: <https://arxiv.org/pdf/2402.08564.pdf>.
- Hart and Mas-Colell, *A Simple Adaptive Procedure Leading to Correlated Equilibrium*. Author page (links to PDFs): <https://ma.huji.ac.il/hart/abs/adapt.html>.
- Syrgkanis and Tardos, *Composable and Efficient Mechanisms*. Free preprint: <https://arxiv.org/pdf/1211.1325.pdf>.
- Liu et al., *Empirical Analysis of EIP-1559: Transaction Fees, Waiting Time, and Consensus Security*. Free preprint: <https://arxiv.org/pdf/2201.05574.pdf>.
- Azouvi et al., *Base Fee Manipulation in Ethereum’s EIP-1559 Transaction Fee Mechanism*. Open-access proceedings PDF: <https://drops.dagstuhl.de/opus/volltexte/2023/19132/pdf/LIPICs-DISC-2023-6.pdf>.
- Chung, Ferreira, Gafni, Yaish, *EC’24 Tutorial on Transaction Fee Mechanism Design*. Free slides: <https://aviv.yai.sh/images/ec24tutorialtfmsl1.pdf>.

3.1 What prior mechanism-design literature implies

L1: Incentive properties must be evaluated jointly across users and block producers.

Transaction fee mechanism design work (notably Roughgarden’s framework for user incentive compatibility, miner incentive compatibility, and off-chain agreement resistance) emphasizes that no practical mechanism simultaneously optimizes all dimensions. For this audit, the implication is direct: the relevant benchmark is not “is this truthful/ESS?” but “which deviations become unprofitable enough under realistic producer behavior?”

L2: EIP-1559 is a hybrid benchmark, not a pure alternative. Economic analyses of EIP-1559 show that algorithmic base-fee updates improve estimation and reduce some pathologies, but priority bidding remains an auction. The comparison point for this proposal should therefore be “hybrid with different privacy and complexity trade-offs,” not “auction vs non-auction.”

L3: Fee-only security and producer dynamics matter in the long run. Classic Bitcoin fee-market work (e.g., Carlsten et al. on instability without block rewards) shows that producer incentives can dominate user-side mechanism intent. For this audit, that reinforces the importance of miner manipulation bounds and explicit anti-collusion assumptions.

L4: Empirical post-1559 evidence shows policy and MEV layers can dominate fee theory. Measurements on Ethereum after EIP-1559 indicate that wallet policy, tip dynamics, and MEV-related ordering incentives can materially affect inclusion outcomes beyond the nominal fee rule. Implication: a Zcash dynamic-fee mechanism needs explicit assumptions about miner selection and relay policy convergence, not only a fee estimator.

L5: Multi-dimensional fee markets suggest being explicit about “the resource being priced.” Work by Chitra, Angeris, and coauthors frames fee markets as pricing one or more underlying resource constraints (bytes, compute, verification cost) and derives update rules from an optimization viewpoint, including dynamic-pricing rules with robustness guarantees under misspecification. For this proposal, the key implication is that the synthetic-fill model and the choice

of “per-action” as the priced unit are not merely accounting details: they define the resource model. This strengthens the recommendation to treat the synthetic model (and potentially alternative resource units) as first-class, and it motivates exploring utilization-driven backstops that are interpretable as pricing congestion of a defined resource.

L6: Collusion limits should be treated as constraints, not edge cases. Recent work by Gafni, Yaish, and collaborators argues that strong collusion resistance is structurally difficult in transaction-fee settings with strategic block producers. For this audit, that implies we should target explicit manipulation-cost bounds and monitoring regimes, rather than relying on an implicit “miners will compete away manipulation” assumption.

L6a: OCA-proofness and SCP are not interchangeable. The barriers paper shows that SCP is strictly stronger than OCA-proofness, so mechanism claims must state which notion is being targeted. For this proposal, this matters because “collusion resistance” can otherwise be overstated: satisfying an OCA-style argument does not automatically imply SCP-level guarantees.

L6b: Deterministic DSIC+MMIC+OCA compatibility is severely constrained. For deterministic single-item mechanisms, Gafni–Yaish characterize DSIC+1-OCA and MMIC+1-OCA classes and show their intersection with DSIC+MMIC+1-OCA is only the trivial mechanism. The constructive implication is to choose an explicit point on the frontier (for example MMIC+OCA with BIC/learning guarantees) instead of aiming for all desiderata simultaneously.

L6c: Randomization helps but does not remove hard barriers. The same line of work gives worst-case limits for randomized mechanisms under DSIC+MMIC+OCA constraints (including nontrivial efficiency barriers). So, introducing randomization or tie-breaking noise should be viewed as a robustness tool, not as a complete escape from impossibility.

L6d: Discretization is a mechanism-design relaxation, not only a privacy choice. Discrete & Bayesian TFM results show that discretizing bids can recover strong properties and positive revenue in regimes where continuous-bid impossibility results bite. This is directly relevant here: fee bucketing is not only a privacy heuristic, it is also a structural restriction that can improve strategic robustness if paired with well-specified reserve/burn logic.

3.2 What scarce-resource pricing literature implies

R1: “What is priced” should be the scarce resource vector, not historical paid fees. The multidimensional pricing literature models each transaction by a resource-consumption vector and prices those resources directly. For this proposal, the key implication is that a median of paid fees is only an indirect proxy for scarcity; the mechanism should explicitly define which resource(s) are scarce (for example bytes, ZIP-317 actions, verification cost, or a vector of these).

R2: The dual formulation gives an observable update signal. In that framework, price updates use the per-resource demand-minus-supply signal (a gradient/subgradient of the dual objective), which is observable from included blocks. This gives a concrete path to a utilization-sensitive backstop without requiring wallets to first bid higher fees.

R3: Regret guarantees are available, and lower bounds are known. For both additive and multiplicative updates, adversarial-regime average regret scales as $O(1/\sqrt{T})$, with matching lower-bound order $\Omega(1/\sqrt{T})$. This gives a realistic target: do not promise rapid pointwise convergence in all regimes; target provable low average regret and bounded lag.

R4: One-dimensional EIP-1559-style updates are a special case. The multidimensional update view contains one-resource EIP-1559-like dynamics as a special case. This supports a hybrid strategy here: keep low-entropy fee buckets for privacy, while driving the bucket baseline with a resource-mismatch update rule.

R5: Resource basis choice is a first-order governance decision. If resources are mis-specified, dynamic pricing can converge to the “wrong” objective even with low regret. So selecting and validating the resource basis (including possible combined resources) should be treated as core protocol design, not implementation detail.

3.3 What learning-theory literature implies

L7: ESS is usually not the primary outcome concept in repeated heterogeneous markets. ESS is a symmetric-population concept, while real fee markets are repeated, role-asymmetric, and partially observed. Learning-based solution concepts (coarse correlated equilibrium under no-regret learning) are typically a better fit for long-run behavior.

L8: If agents exhibit vanishing external regret, time-average play approaches coarse correlated equilibrium (CCE). The core theorem from no-regret learning implies that if each player’s average regret goes to zero, empirical play converges to an ε -CCE with $\varepsilon \rightarrow 0$. This gives testable predictions without requiring full rational-expectations equilibrium.

L9: Welfare guarantees can transfer from equilibrium to learning only under additional structure. Smoothness-style arguments can carry Price-of-Anarchy-type bounds to no-regret dynamics, but only when the mechanism satisfies suitable smoothness conditions. For this mechanism, proving or rejecting such conditions is a concrete research target.

Synthesis for this audit. These papers strengthen five conclusions:

1. claims should be framed as *learning-robust* (no-regret/CCE) before claiming ESS-like stability;
2. design objectives must be explicit about which collusion notion is targeted (OCA vs SCP);
3. deterministic “all-properties-at-once” targets are infeasible, so the intended compromise frontier must be named up front;
4. the priced object should be a clearly specified scarce-resource model, not only observed paid fees;
5. empirical validation should estimate actual learning rules and producer policies, not only simulate users against a fixed inclusion oracle.

4 Mechanism Under Review

4.1 High-level design goals (as stated)

The proposal aims to:

1. avoid fee choices that leak information that can segment the user base (privacy constraint),
2. use only already-public information,
3. keep fees easy to explain and low under normal conditions,
4. ensure “normal” transactions confirm quickly when uncongested,
5. respond quickly to congestion and slowly return to normal prices,
6. make griefing (preventing legitimate confirmations) expensive,
7. keep consensus rules simple while enabling a free market with a stable strategy.

4.2 Concrete estimator (as implemented in the lab)

The lab materials and implementation behavior define an on-chain-data estimator for a *recommended* ZIP-317 marginal fee. Let $B = 50$ and let the tip buffer be $b = 5$ blocks. At height h , define a lookback window

$$W(h) = \{h - b - B + 1, \dots, h - b\}.$$

Notation. We use f for the fee floor (in zats per conventional action), $S_{\max}$ for the consensus maximum block size (bytes), and S_i for observed block size. The lab’s sizing uses decimal kilobytes: $x^{\text{KB}} := \lfloor x/1000 \rfloor$.

Per-transaction primitives (ZIP-317). Each transaction reports a *conventional action count* $a \in \mathbb{N}$ (called `conventional_actions` in the lab data). The transaction fee paid is $F \in \mathbb{N}$ (zatoshi). Define the per-action fee

$$p := \left\lfloor \frac{F}{a} \right\rfloor.$$

Synthetic transactions (“always congested”). For each block $i \in W(h)$, let S_i be its size and let $S_{\max}$ be the consensus maximum block size (both in bytes). The lab converts to kilobytes as $S^{\text{KB}} := \lfloor S/1000 \rfloor$. Let $\tilde{s}_i$ be the *median* transaction size (in KB) among transactions in block i . The lab computes a synthetic-transaction count

$$k_i := \left\lfloor \frac{S_{\max}^{\text{KB}} - S_i^{\text{KB}}}{\tilde{s}_i} \right\rfloor,$$

and appends k_i synthetic transactions with per-action fee equal to a fixed *fee floor* f (the lab uses $f = 10$ zats as an illustrative value).

Block median and window median. For each block i , let $\mathcal{P}_i$ be the multiset of per-action fees of its real transactions, and let $\mathcal{P}'_i := \mathcal{P}_i \cup \{f \text{ repeated } k_i\}$. Define the block median

$$m_i := \text{median}(\mathcal{P}'_i).$$

Define the window median

$$M := \text{median}(\{m_i : i \in W(h)\}).$$

Rounding (privacy/UX constraint). The estimator rounds M to a power of ten via a most-significant-digit rule: for $M \geq f$, let 10^n be the largest power of ten not exceeding M . Then

$$\text{Round}_{10}(M) := \begin{cases} 10^n, & M < 5 \cdot 10^n, \\ 10^{n+1}, & M \geq 5 \cdot 10^n. \end{cases}$$

Finally the recommended marginal fee is $\hat{m} := \max(f, \text{Round}_{10}(M))$.

Fast lane (priority tier). The lab defines a fast-lane recommendation as

$$\hat{m}_{\text{fast}} := 10 \cdot \hat{m}.$$

The proposal text describes opening this “fast lane” only under extreme congestion (informally: when no synthetic transactions “fit” in the lookback window).

Parameter sensitivity (qualitative).

Parameter	Primary effect	Upside	Main risk
f (fee floor)	Baseline marginal fee in uncongested regime	Raises spam cost immediately	If too high, harms low-value usage; if too low, weak DoS resistance
B (lookback)	Estimator smoothing horizon	Lower variance, less manipulation by short bursts	Slower response to shocks; larger underpayment windows
b (tip buffer)	Reorg robustness in sampled data	Fewer reorg-induced fee oscillations	Additional lag in reacting to true congestion changes
Fee alphabet	Granularity of bids	Lower fee entropy (privacy), simpler UX	Larger bucket cliffs and boundary manipulation incentives
Fast-lane multiplier	Priority spread under congestion	Potential emergency valve for urgency traffic	Poor fit to heterogeneous demand if fixed; can leak urgency/wealth

4.3 Intended consensus changes (as described)

The proposal suggests a path from pure monitoring to policy and then (optionally) to consensus constraints. The text sketches consensus rules along the lines of:

- transactions must have expiry height within the lookback horizon,
- per-action fee choices are restricted to powers of ten,
- per-action fee must be above a defined floor.

This report treats the estimator as the core mechanism and evaluates which of these constraints are required to make it incentive-robust.

5 Threat Model and Participants

Participants.

- **Users/wallets** choose fees and expiry heights to maximize confirmation probability subject to cost and privacy constraints.
- **Relay nodes** (policy) decide what to accept and forward (e.g., minimum fees, bucketing enforcement).
- **Miners** choose transaction selection policies to maximize revenue (fees plus block subsidy) and may coordinate.
- **Adversaries** may be profit-motivated (miner cartels, MEV-like extractors) or griefing-motivated (sandblasting/spam).

Adversarial capabilities assumed. Attackers can generate arbitrarily many valid transactions, choose fees, and coordinate timing; miners can include/exclude transactions and can create self-transactions. We assume attackers cannot break consensus cryptography but can exploit all policy/consensus degrees of freedom.

6 Economic Analysis and Findings

6.1 What kind of mechanism is this?

The proposal is *not* itself a market-clearing rule baked into consensus (unlike EIP-1559’s base fee update). Instead it is best modeled as a **posted-price estimator** (a recommended marginal fee) coupled with a **restricted bid space** (powers of ten) and optionally a **short time-to-live** (expiry-height bound). Whether this becomes a stable “fee market” depends primarily on relay policy adoption and miner selection behavior.

6.2 F0: Policy convergence is part of the mechanism

Finding (F0). In the policy-first deployment path, the estimator does not bind behavior unless relay nodes and miners largely converge on a compatible mempool/inclusion policy. If a meaningful fraction of relays accept lower-fee or non-bucketed transactions (or if miners include them anyway), the system can quickly revert to first-price dynamics with heterogeneous wallet heuristics.

Recommendation. Treat relay policy convergence as a first-class design requirement: specify the expected node policy (fee alphabet, minimums, expiry constraints) and measure its adoption before relying on the estimator for spam resistance or UX guarantees.

6.3 Fixed points and “stable strategies” (qualitative)

Ignoring rounding, the estimator is approximately a feedback function from the distribution of per-action fees in recent blocks to the next recommended marginal fee. If most transactions pay the current conventional fee (ZIP-317), then p concentrates at the current marginal fee, and the estimator has a natural fixed point: it recommends the same marginal fee again. This is the core argument for a stable “everyone pays the recommendation” strategy.

However: stability hinges on a spam-resistance lever. If the network is uncongested, synthetic transactions dominate the per-block median whenever blocks are less than roughly half full (in terms of synthetic-count construction). In that regime, the estimator collapses toward the floor f . That is desirable for low fees, but it implies that the **fee floor is the primary anti-spam parameter** in low-demand regimes.

6.4 F1: Congestion does not mechanically raise fees

Finding (F1). Under this mechanism, fees only rise if *included* transactions pay higher per-action fees. High congestion alone does not force an increase. Therefore a spammer can, in principle, keep blocks full using transactions that pay (approximately) the fee floor, without triggering any endogenous fee escalation.

Why this matters. This is materially different from EIP-1559, where the base fee increases mechanically with utilization even if every transaction pays the minimum. In the comparable-based estimator, if miners are willing to include large volumes of floor-fee traffic (e.g., because there is little competing demand), then the system has no automatic “pressure valve” other than external intervention (raising the floor) or miner behavior (preferring higher-fee traffic).

Implication. The proposal’s goal “griefing becomes expensive” is only satisfied to the extent that either:

1. the fee floor f is set high enough that sustained throughput-level spam is costly, or
2. there exists a reliable priority mechanism that legitimate users can use to outbid spam at acceptable cost, and miners actually honor that priority.

6.5 F2: Fee-floor selection becomes a governance/security parameter

Finding (F2). Because the estimator can sit at (or near) the floor for extended periods (by design), the floor f becomes a de facto *security parameter* rather than a mere UX default. Setting f too low reintroduces the core vulnerability of fixed low fees described in ZIP-313: cheap denial-of-service.

Recommendation. Define an explicit policy for selecting and updating f (and who controls it), ideally with an automatic component tied to observable congestion signals. If governance must update f manually, incorporate “time-to-respond” into the threat model.

6.6 F3: Miner manipulation of the estimator

Finding (F3). Miners influence the estimator because it is computed from included transactions. A miner (or cartel) can attempt to raise future recommended fees by:

1. selectively including higher-fee transactions (and excluding lower-fee ones),
2. creating self-transactions that pay high fees (“wash fees”) to move medians, especially near rounding thresholds.

Cost of manipulation. The median-of-medians structure and the synthetic-floor construction increase the cost of manipulation relative to naive averages: outliers do not move medians, and in low-utilization blocks the synthetic floor pins m_i to f unless real traffic occupies a majority of the synthetic-count model. However, manipulation may still be profitable if miners can coordinate and if they internalize the future revenue externality.

ZIP-235 interaction. If ZIP-235 (or any mechanism) burns a fixed fraction of fees, miner self-transactions become meaningfully costly: even conditional on mining the block, the burned share is a real cost, and conditional on *not* mining the block, the full fee is lost. This is a strong mitigation against “wash fee” manipulation. Without burning, wash-fee manipulation can be close to costless for a miner who expects to mine the block.

6.7 F4: Coarse rounding creates discontinuities and boundary incentives

Finding (F4). Restricting fees to powers of ten is privacy- and UX-friendly (few buckets), and it eliminates the “epsilon outbid” arms race of continuous first-price bidding. However, it also creates large discontinuities: when M crosses $5 \cdot 10^n$, the recommended marginal fee jumps by $10\times$.

Consequences.

- **Boundary manipulation:** miners or large senders have concentrated incentive to push M just over a threshold.
- **Oscillation risk:** if demand is price-sensitive, the system can flip between buckets (a form of limit cycle), particularly if policy enforces the recommendation tightly.
- **Inefficiency:** users whose true willingness-to-pay lies between buckets must overpay by up to $10\times$ to buy priority.

Recommendation. Consider a slightly richer but still low-entropy fee alphabet, e.g., $\{1, 2, 5\} \cdot 10^n$, or rounding *down* only. This preserves privacy goals while reducing discontinuity magnitude.

6.8 F5: “Fast lane” effectiveness depends on miner selection rules

Finding (F5). The proposal’s fast lane is defined as paying $10\times$ the marginal fee. Under the ZIP-317 recommended block template algorithm, prioritization is probabilistic and capped by `weight_ratio_cap` (default 4). As a result, a $10\times$ fee may not translate into a $10\times$ inclusion advantage under default mining software behavior.

Recommendation. If a fast lane is intended to be an operationally reliable emergency valve, specify (and ideally standardize) miner selection behavior during fast-lane conditions. Otherwise the fast lane should be framed as “increases odds” rather than “goes first”.

6.9 F6: Lookback lag and attack windows

Finding (F6). Using a 50-block lookback plus a 5-block buffer improves reorg robustness and smooths noise, but it also introduces lag. In rapid demand spikes (or the onset of a spam attack), many users will initially underpay relative to the new congestion level, and the estimator will only adjust after sufficient high-fee blocks enter the window.

Recommendation. If “respond quickly to congestion” is a hard requirement, consider (i) a shorter or exponentially weighted window, or (ii) a utilization-driven update rule (EIP-1559-like) that does not depend on users first choosing higher fees.

6.10 F7: Synthetic-count model is a mechanism lever and an attack surface

Finding (F7). The synthetic transaction construction effectively embeds a target utilization concept (a block needs to be “more than half full” in the synthetic-count sense for real fees to dominate the median). This is powerful but must be specified precisely:

- what size metric is used (median vs average; per-block vs window),
- how to handle edge cases (small transactions, near-zero medians),
- whether the synthetic model should be based on bytes, actions, or verification cost.

Recommendation. Treat the synthetic model as a first-class part of the spec. Consider action-weighted or byte-weighted medians (median over *block resource units*) to reduce sensitivity to transaction sizing quirks and to better match congestion costs.

6.11 F8: Sharp transition at the “synthetic displacement” boundary

Finding (F8). Because synthetic transactions are included at the fee floor and then a *median* is taken, the recommended fee can remain pinned to the floor over a wide range of low-to-moderate utilization, and then change abruptly once real-fee observations dominate the synthetic construction. This is a deliberate design choice (stable, low fees when uncongested), but it implies a **phase transition** rather than a smooth response curve.

Why this matters. At the boundary between buckets (e.g., around $5 \cdot 10^n$), small changes in traffic composition can trigger a $10\times$ jump in the recommended marginal fee. If demand is price-sensitive, the system may exhibit **hysteresis/oscillation**: the fee jumps up, marginal users drop out, the median falls back, and the system flips again. Whether this occurs depends on the distribution of user urgency and on relay/miner policies.

Recommendation. Simulate the transition regime explicitly (agent-based or trace-driven simulation) and consider mechanisms that reduce boundary sharpness (richer fee alphabet, rounding-down only, or a utilization-driven backstop that raises cost smoothly under sustained full blocks).

6.12 F9: The estimator prices outcomes, not scarce resources

Finding (F9). The current estimator updates from *paid fee observations* rather than direct measurements of resource mismatch. This can work in stable regimes, but it weakens response under strategic shading or floor-fee saturation: scarcity can rise while paid-fee medians remain pinned.

Why this matters. Resource-pricing results suggest that robust dynamic updates should follow a measurable demand-minus-supply signal for the constrained resource(s). A fee-median-only signal can lag or fail exactly in the regimes where anti-spam and congestion control are most needed.

Recommendation. Keep the low-entropy fee alphabet for privacy, but drive the baseline tier with a resource-mismatch backstop (for example, additive or multiplicative updates on a defined resource basis), and use fee medians as a secondary smoothing signal rather than the sole control signal.

6.13 Claim-to-evidence matrix

ID	Claim	Current support	High-value next test
F0	Policy convergence is required for estimator to bind behavior	Mechanism logic	Measure relay/miner policy adoption and heterogeneity
F1	Congestion alone does not force fee increases	Mechanism logic	Replay full-block floor-fee traces and observe non-escalation
F2	Floor acts as security parameter	ZIP-313 + mechanism logic	Estimate minimum floor/backstop for uneconomic sandblasting
F3	Miner manipulation is possible but costed by medians and burning	Mechanism logic + ZIP-235 intuition	Bound manipulation cost vs hashrate share and burn fraction
F4	Power-of-ten rounding causes cliff effects	Deterministic from rounding rule	Simulate threshold crossing and bucket-cycling frequency
F5	Fast lane may not deliver proportional priority	ZIP-317 selection cap logic	Estimate inclusion-probability gain for each tier under load
F6	50+5 window introduces lag during shocks	Deterministic from estimator horizon	Measure response half-life and underpayment depth on spikes
F7	Synthetic-count model encodes a hidden resource model	Mechanism logic + multi-dimensional fee literature	Compare bytes/action/verification-cost synthetic models
F8	Synthetic displacement can create phase transitions/hysteresis	Mechanism logic	Agent-based simulation with price-sensitive demand
F9	Fee-median signal can miss direct scarcity changes	Mechanism logic + resource-pricing literature	Compare fee-median-only vs resource-mismatch backstop on shocks

7 Privacy Considerations

Fee entropy reduction (strength). Restricting per-action fees to a small, discrete alphabet (powers of ten) substantially reduces fee-based fingerprinting relative to continuous fee choices. This is aligned with Zcash’s privacy goals and is a primary differentiator versus EIP-1559-style continuous tip markets.

Boundary timing leakage (concern). When the recommended fee changes infrequently but discontinuously, transactions near a transition can become timing signals (“paid the old bucket just before the flip”). The 5-block buffer improves reorg robustness but does not remove this effect.

Fast-lane segmentation (concern). If a fast lane is active, the fee becomes a binary attribute (normal vs $10\times$). This can leak information about urgency/wealth and may enable user segmentation

in exactly the way the mechanism is trying to avoid. If a fast lane is retained, it should be explicitly justified as an acceptable privacy trade-off, and wallets should make its use predictable (to avoid creating extra entropy via wallet-specific heuristics).

Wallet fingerprinting (concern). If different wallets and services implement the estimator differently, or choose different “fast-lane” rules, fee selection becomes a wallet-type fingerprint. This pushes toward standardizing policy and clearly documenting the recommended behavior.

8 Comparisons and Design Space

Versus fixed fees (ZIP-317 status quo). In the common uncongested regime, the mechanism reduces to “a fixed fee floor with extra steps,” because the estimator intentionally pins to f . Its main advantage over a purely fixed fee is *conditional adaptability*: if real transaction fees rise in included blocks, the recommendation can rise too.

Versus EIP-1559. EIP-1559 implements a utilization-driven base fee update in consensus (plus a tip market). The Shielded Labs approach keeps consensus simpler and improves privacy by discretizing fees, but it inherits a key limitation: without a utilization-driven backstop, sustained full blocks at low fees need not cause price to rise.

Hybrid directions. Two hybrid directions appear especially compatible with the proposal’s goals:

- **Utilization-driven floor/backstop:** add a slow, monotone increase in the floor when blocks remain full for long enough, while keeping the discrete fee alphabet for privacy.
- **External-cost anchoring for f :** periodically adjust f based on a slow-moving on-chain signal (e.g., mining difficulty as a rough proxy for costs), while retaining the short-horizon median estimator for congestion response.

Quick comparison table.

Dimension	EIP-1559-like	Comparable-median proposal
Price update	Utilization-driven (consensus)	Median of included fees (policy-first)
Priority market	Continuous tips	Discrete buckets / tiers
Privacy (fee entropy)	Poorer	Better
Shock response	Smooth / fast	Lagged; phase transitions
Manipulation cost	High via burning	Moderate; burn/backstop critical

9 Recommendations (Summary)

Recommendations are split into actions that are deployable in a policy-first pilot versus actions that should gate consensus-level commitments.

9.1 Deploy-now actions (policy-first)

1. **D1 (Floor is security):** define an explicit floor rule (not an ad hoc knob) and publish governance/update cadence.

2. **D2 (Utilization backstop):** add a monotone utilization-triggered floor adjustment in relay policy.
3. **D3 (Priority tiers):** replace single $10\times$ fast lane with a small tier set (e.g., $1\times, 2\times, 5\times$).
4. **D4 (Policy convergence):** standardize mempool and block-template policy, then publish adoption telemetry.
5. **D5 (Privacy accounting):** publish entropy and segmentation metrics for each fee-policy release.
6. **D6 (Property frontier):** explicitly state the targeted property mix (for example MMIC+OCA with learning-robust behavior), rather than claiming simultaneous DSIC/MMIC/collusion-resistance.
7. **D7 (Resource signal):** publish a policy-level resource mismatch signal (even one-dimensional at first) and use it as a backstop input for baseline fee updates.

9.2 Research-gated actions (before consensus lock-in)

1. **G1 (Synthetic model spec):** choose and justify the resource model (bytes/actions/verification cost) with empirical fit.
2. **G2 (Rounding design):** validate whether powers-of-ten should be replaced by $\{1, 2, 5\} \cdot 10^n$ or directional rounding.
3. **G3 (Formal repeated-game model):** specify strategy spaces/information and evaluate outcomes under no-regret learning.
4. **G4 (Manipulation bounds):** produce explicit attacker-cost bounds parameterized by hashrate share and burn fraction.
5. **G5 (Consensus migration criteria):** define quantitative thresholds that must be met before any consensus-level fee constraints.
6. **G6 (Burned-reserve form):** evaluate whether reserve-and-burn structure (suggested by OCA characterizations) can be implemented with privacy-preserving discrete tiers.
7. **G7 (Regret-accounted controller):** specify update family, step-size policy, and admissible bounds (B, M, ϵ) , then report regret diagnostics as first-class release criteria.

10 On Nate’s “Three Classes” Fee-Market Taxonomy

The following heuristic taxonomy was proposed informally:

1. “Wild West” $\approx$ first-price bidding with no stable equilibrium,
2. EIP-1559, moving the fee market inside consensus,
3. this proposal: simple consensus + free market + evolutionarily stable strategy.

On the “ESS” claim. An Evolutionarily Stable Strategy (ESS) is a precise notion from evolutionary game theory. In a symmetric two-player game with payoff function $u(\cdot, \cdot)$, a strategy σ^* is an ESS if for every mutant strategy $\sigma \neq \sigma^*$, either

$$u(\sigma^*, \sigma^*) > u(\sigma, \sigma^*),$$

or

$$u(\sigma^*, \sigma^*) = u(\sigma, \sigma^*) \text{ and } u(\sigma^*, \sigma) > u(\sigma, \sigma).$$

To meaningfully assert an ESS here, the proposal must define the strategy space (fee choices, expiry choices, miner inclusion policies), the information structure, and the payoff function, and then show that a particular behavior (e.g., “pay the recommended bucket”) is robust to deviations. In practice, a no-regret/CCE benchmark is likely the more operational equilibrium concept for this setting.

Assessment. The direction is useful but the claims are too coarse:

- First-price auctions do have equilibria under standard assumptions, but real-world mempools create repeated, latency-sensitive games where stable bidding heuristics can break down.
- EIP-1559 moves the *base fee* update into consensus but retains an off-chain priority fee (tip) market.
- The Shielded Labs proposal is best viewed as a *posted-price estimator with a discrete bid space*. It can create a practically stable “pay the recommendation” strategy *if* relay policy is widely adopted and if the anti-spam floor/backstop is strong. Without that, it can degenerate back into first-price dynamics.

Suggested refinement. A more mechanism-design-oriented classification is:

1. **Auction-dominant:** continuous bids, miner revenue maximization, equilibrium depends on latency and mempool rules.
2. **Base-fee dominant (consensus):** utilization-driven base fee, tips for priority (EIP-1559-like).
3. **Posted-price / schedule:** deterministic (or slowly varying) fee schedule with limited priority tiers; stability comes from restricting profitable deviations.
4. **Governance-fixed (baseline):** fixed marginal fees (or a fixed floor) chosen by governance; simple and predictable but vulnerable if set too low or too high for long periods.

11 Learning Dynamics and No-Regret Interpretation

Why this matters for current claims. Given the repeated and partially observed nature of fee selection, the right near-term target is not a strict ESS proof but a statement of robustness under plausible learning dynamics.

External regret benchmark. Let player i choose actions a_i^t over rounds $t = 1, \dots, T$ with utility $u_i(a_i^t, a_{-i}^t)$. Define external regret

$$\text{Regret}_i(T) = \max_{a'_i} \sum_{t=1}^T u_i(a'_i, a_{-i}^t) - \sum_{t=1}^T u_i(a_i^t, a_{-i}^t).$$

If $\text{Regret}_i(T)/T \rightarrow 0$ for all players, empirical play converges to an ε -coarse correlated equilibrium with $\varepsilon \rightarrow 0$.

Interpretation for this mechanism. If wallets/users/miners are approximately no-regret learners:

- we can claim “no fixed unilateral deviation improves long-run average payoff by more than ε ”;
- we cannot claim convergence to a unique fee bucket or strict Nash equilibrium;
- persistent cycling across buckets can still occur while average regret remains low.

Empirical studies that can be run on existing systems.

1. **Regret backtesting on historical traces:** infer sender/wallet policy classes and compute realized regret against best fixed bucket/tip policies in hindsight.
2. **Learning-rule model selection:** fit and compare candidate updates (best response, exponentially weighted updates, bandit-style exploration) using out-of-sample likelihood.
3. **Shock-response identification:** around exogenous congestion spikes, estimate reaction speed, overshoot, and decay for fee updates.
4. **Producer-policy inference:** estimate miner inclusion scoring from observed blocks to determine whether users face stable or drifting objectives.
5. **Counterfactual replay:** replay real mempool traces under this mechanism with calibrated learners to estimate delay, welfare, and manipulation cost.
6. **Controlled testnet experiments:** deploy known bot learners against configurable producer policies to map which learner signatures match field data.

Operational signatures (how to distinguish learning families). In data, no-regret dynamics should show decreasing average external regret with horizon, potentially persistent mixed support over several fee buckets, and muted one-shot best-response jumps. By contrast, myopic best-response dynamics should show sharper bucket-hopping and higher exploitable predictability after shocks. This distinction is testable and should be part of acceptance criteria before using ESS-like language in public claims.

What stronger results become possible. If data supports low-regret behavior, the audit can be strengthened from “possible stable heuristic” to:

1. a measurable ε -CCE robustness statement for deployed policy; and
2. conditional welfare bounds, if smoothness-like inequalities can be established for the discrete-bucket mechanism.

12 Experiment Specifications (Next Quarter)

1. **E1 Policy convergence.** Dataset: relay policy snapshots + mined blocks for policy-first clients. Metric: share of blocks consistent with target fee policy. Decision threshold: proceed only if $> 85\%$ for 4 consecutive windows. Owner: node/infra team.

2. **E2 Shock response.** Dataset: historical congestion shocks + replay harness. Metric: response half-life of $\hat{m}$ and underpayment depth. Decision threshold: median half-life < 25 blocks and P95 delay no worse than $+20\%$ vs baseline. Owner: wallet + research.
3. **E3 Tier effectiveness.** Dataset: mempool replays with $1\times, 2\times, 5\times$ tiers. Metric: inclusion-probability gain by tier. Decision threshold: gains must be monotone and each step must improve inclusion by at least $1.5\times$ under load. Owner: mining + wallet.
4. **E4 Manipulation bounds.** Dataset: adversarial replay sweeping miner share α and burn fraction β . Metric: cost to push one bucket upward for one window. Decision threshold: attacker cost must exceed expected extractable gain by at least $2\times$ for $\alpha \leq 0.3$. Owner: research/econ.
5. **E5 Transition stability.** Dataset: agent-based simulation with price-sensitive demand. Metric: bucket-switch frequency and dwell time. Decision threshold: fewer than 1 switch per 100 blocks outside shock windows. Owner: research/econ.
6. **E6 Synthetic model choice.** Dataset: historical blocks with bytes/actions/verification-cost proxies. Metric: out-of-sample fit for congestion and delay. Decision threshold: choose model with best delay prediction and no worse than $+10\%$ privacy entropy vs baseline. Owner: protocol research.
7. **E7 Privacy accounting.** Dataset: aggregated wallet telemetry on bucket/tier usage. Metric: fee entropy and fast-lane segmentation lift. Decision threshold: entropy reduction must be material while segmentation stays below the agreed privacy budget. Owner: privacy team.
8. **E8 Learning robustness.** Dataset: trace-driven policy inference + learner fitting. Metric: empirical ε for no-regret/CCE diagnostics. Decision threshold: promote learning-robust claims only if ε is small and stable over rolling windows. Owner: research/econ.
9. **E9 Learning-rule discrimination.** Dataset: matched sender cohorts around congestion shocks. Metric: out-of-sample log-likelihood gap between no-regret-style models and myopic best response. Decision threshold: claim “no-regret plausible” only if no-regret family wins by a pre-registered margin across at least three shock regimes. Owner: research/econ.
10. **E10 Collusion frontier audit.** Dataset: adversarial replay with miner-user coalitions and side-contract channels. Metric: coalition gain over protocol outcome under OCA-style and SCP-style tests. Decision threshold: publish explicit residual collusion surface and block consensus migration unless high-risk regions are bounded by policy mitigations. Owner: research/econ + protocol.
11. **E11 Resource-basis ablation.** Dataset: same traffic replay under candidate resource maps (actions-only, bytes+actions, bytes+actions+verification proxy). Metric: welfare, delay, and privacy entropy under matched controller settings. Decision threshold: choose basis on Pareto frontier with no dominated candidate on all three dimensions. Owner: protocol research.
12. **E12 Controller benchmark.** Dataset: trace-driven replay comparing three controllers (fee-median, utilization-backstop, hybrid). Metric: average regret proxy, shock-recovery time, and bucket oscillation rate. Decision threshold: adopt hybrid only if it weakly dominates fee-median on all three metrics across stress scenarios. Owner: research/econ + wallet.

13 Falsification Criteria

Purpose. To keep the audit decision-useful, each major claim should have a practical falsifier.

1. **Anti-spam falsifier:** if sustained floor-fee saturation can keep P95 confirmation delay elevated for legitimate traffic without triggering backstop escalation, claim F1/F2 mitigation is not sufficient.
2. **Priority-tier falsifier:** if higher tiers do not deliver monotone inclusion gains under load, fast-lane/tiered urgency claims are invalid in current miner policy.
3. **Stability falsifier:** if bucket-switch frequency remains high outside shocks, stability claims are overstated.
4. **Learning falsifier:** if fitted learning dynamics fail no-regret diagnostics (or are better explained by unstable myopic responses), ESS-like framing should be removed from summaries.
5. **Manipulation falsifier:** if estimated manipulation cost at realistic miner share is below expected extractable gain, manipulation resistance claims should be downgraded.

14 Conclusion and Suggested Next Steps

The proposal’s strongest contribution is a privacy-compatible fee space (low entropy) with a deployable path (policy-first). Its primary mechanism-design risk is that it can permit sustained full-block conditions without endogenous price escalation unless the floor/backstop is strong and widely enforced.

Pilot decision set (policy mode).

1. **Decision A (fee alphabet):** pilot $\{1, 2, 5\} \cdot 10^n$ with rounding down to reduce cliff effects while preserving low entropy.
2. **Decision B (backstop trigger):** increase the floor by one tier if block fullness exceeds 85% for three consecutive windows; decay by one tier only after twelve low-fullness windows.
3. **Decision C (priority tiers):** replace binary fast lane with $1\times, 2\times, 5\times$ tiers and publish wallet defaults tied to urgency classes.

Suggested next steps.

1. Specify a concrete utilization-driven backstop (even if initially only in policy) and evaluate its privacy impact.
2. Standardize relay/miner policy assumptions and measure convergence in the field.
3. Run trace-driven backtests to estimate regret and identify whether users/miners behave like no-regret learners.
4. Iterate on the fee alphabet and priority tiers to reduce cliff effects while preserving low entropy.
5. Publish an explicit property target (which collusion notion and which incentive notion) and keep claims aligned with that target.

6. Finalize and justify the scarce-resource basis being priced, then benchmark controller performance against that basis.